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ARTICLE

Do natural history data predict the movement ecology of fishes in Lake Ontario streams?

Ivan J. Dolinsek, Robert L. McLaughlin, James W.A. Grant, Lisa M. O'Connor, and Thomas C. Pratt

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Abstract: Little is known about the movements of most stream fishes, so fisheries managers often rely on natural history data from the literature to make management decisions. Observations of over 15 000 individuals from 37 species across 3 years were used to evaluate four aspects of the reliability of literature data for predicting the movement behaviour of stream fishes: (*i*) water temperature when fish enter streams; (*ii*) reasons for moving into the streams; (*iii*) stream residence times of migrants; and (*iv*) relative use of lake and stream habitats. Comparisons of our data for arrival times in the streams, water temperature at arrival, and time spent in the streams were highly correlated with literature data, whereas relative use of the lake was not. Further, our detailed data revealed two novel findings: (1) in many species juveniles were also moving into streams, even in those species where adults were clearly spawning in the streams; and (2) adult-sized individuals were moving into streams for nonreproductive purposes. Our results suggest that fishery managers can confidently use natural history information to gain general insights into the movement ecology of fishes, but should also recognize that this information remains incomplete in important ways.

Résumé : Les connaissances sur les déplacements de la plupart des poissons de cours d'eau sont très limitées, de sorte que les gestionnaires des pêches ont souvent recours à des données d'histoire naturelle tirée de la littérature pour éclairer leurs décisions de gestion. Des observations sur plus de 15 000 individus de 37 espèces recueillies pendant 3 ans ont été utilisées pour évaluer quatre aspects de la fiabilité des données tirées de la littérature pour ce qui est de prédire le comportement de déplacement des poissons de cours d'eau, soit (*i*) la température de l'eau quand les poissons entrent dans les cours d'eau, (*ii*) les raisons qui les poussent à entrer dans les cours d'eau, (*iii*) les temps de séjour des migrants dans les cours d'eau et (*iv*) l'utilisation relative d'habitats de lac et de cours d'eau. Nos données sur les temps d'arrivée dans les cours d'eau, la température de l'eau à l'arrivée et le temps passé dans les cours d'eau sont fortement corrélées aux données tirées de la littérature, ce qui n'est pas le cas pour les données sur l'utilisation relative de lacs. En outre, nos données détaillées révèlent deux nouvelles constatations, soit que (1) pour de nombreuses espèces, les juvéniles entrent également dans les cours d'eau, même chez les espèces pour lesquelles il est clair que les adultes frayent dans les cours d'eau et (2) les individus de taille adulte entrent dans les cours d'eau à des fins autres que la reproduction. Nos résultats donnent à penser que les gestionnaires des pêches peuvent se fier aux données sur l'histoire naturelle pour établir une compréhension générale de l'écologie des déplacements des poissons, mais qu'ils doivent également tenir compte du fait que ces renseignements demeurent incomplets à d'importants égards. [Traduit par la Rédaction]

Introduction

Human impacts on the biosphere are pressing scientists to provide information and solutions to help conserve native biodiversity and ecosystem services (Palmer et al. 2004; Venter et al. 2006). To date, most of the focus has been on species declines in terrestrial habitats, particularly in the tropical forests, because these ecosystems are perceived to be in greater peril (Ricciardi and Rasmussen 1999). However, studies have suggested that freshwater fauna are more threatened than terrestrial species (Richter et al. 1997; Ricciardi and Rasmussen 1999). Globally, 21% (n = 1851) of freshwater fishes that have been evaluated by the International Union for Conservation of Nature (IUCN) are considered to be threatened (IUCN 2010). In Canada, 30% of the freshwater and diadromous fishes have been assessed by the Committee on the Status of Endangered Wildlife in Canada (COSEWIC) as being at risk throughout all or parts of their ranges (Hutchings and Festa-Bianchet 2009; COSEWIC 2010).

Successful conservation and recovery plans rely upon the best available knowledge regarding the biology, ecology, and the life history of a species (Abell 2002; Poos et al. 2008). Unfortunately, detailed information on the population structure and life histories of many freshwater fishes is often limited, anecdotal, or unavailable (Abell 2002; Mandrak et al. 2003; Poos et al. 2008). Decision makers are therefore faced with the decision of relying on existing natural history data, at the potential risk of inappropriate management or conservation actions should the existing data be incomplete or inaccurate, or finding additional funding, resources, and time to collect the needed information (Smith and Jones 2007).

Understanding the movement behaviour and habitat use of fishes could be valuable for conservation and recovery plans. Many freshwater fishes require specific habitats to complete the different stages of their life cycles (Lucas and Baras 2001). These habitats are often spatially separated, so individuals may move

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I.J. Dolinsek* and J.W.A. Grant. Biology Department, Concordia University, Montréal, QC H4B 1R6, Canada.
R.L. McLaughlin. Department of Integrative Biology, University of Guelph, Guelph, ON N1G 2W1, Canada.
L.M. O'Connor and T.C. Pratt. Fisheries and Oceans Canada, Great Lakes Laboratory for Fisheries and Aquatic Sciences, 1219 Queen Street E., Sault Ste-Marie, ON P6A 2E5, Canada.
Corresponding author: Ivan J. Dolinsek (e-mail: ivan.dolinsek@umontreal.ca).
*Present address: Department of Biological Sciences, Université de Montréal, C.P. 6128, Succursale Centre-ville, Montréal, QC H3C 3J7, Canada.

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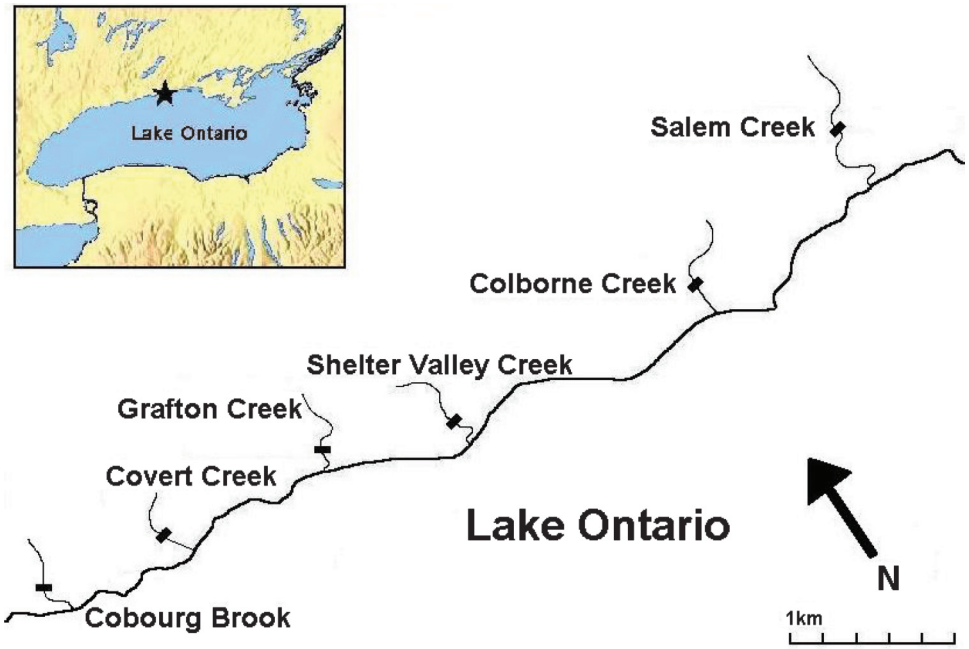
Fig. 1. Map indicating the locations of the study streams and the approximate positions of the first upstream barrier within each stream (black rectangle). Inset map shows the location of the study area (star) in relation to Lake Ontario.



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long distances, often between different tributaries or across bodies of water, to reach suitable habitats to meet their needs. These movements can affect reproductive fitness and help maintain metapopulation dynamics and stabilize fragmented populations via the rescue effect (Wilson et al. 2004; Primack 2008). However, early studies assumed many freshwater fishes were sedentary or exhibited restricted movement (Gerking 1959; Gowan and Fausch 1996), because of imprecise and limited sampling. Results from more recent tracking studies indicate that movement in many populations of stream fishes is more common and more variable among populations and species than previously thought (Lucas and Baras 2001; Rodriguez 2002; Mandrak et al. 2003). Some studies report partial migration, with individuals adopting migration or residency as alternative life history strategies (Kerr et al. 2009; Chapman et al. 2012).

New tracking technologies, such as passive integrated transponder (PIT) tags, create opportunities to track the movements of small fishes more quantitatively, widely, and affordably (Roussel et al. 2000). This technology has been an effective and valuable tool in movement studies at the scale of single streams. When combined with strategically placed antennas, it can facilitate the collection of data on the movement behaviour and habitat use by individuals of different size and life stage (Ombredane et al. 1998). Information on the movement behaviour of many species is often limited, unavailable, or lacks quantitative evidence (Abell 2002; Mandrak et al. 2003; Poos et al. 2008).

Most examinations of movement for freshwater fishes have focussed on species with commercial or recreational value (i.e., walleye, *Sander vitreus*; yellow perch, *Perca flavescens*; and salmonids) (Northcote 1998; Lucas and Baras 2001; Landsman et al. 2011), invasive species (i.e., sea lamprey, *Petromyzon marinus*) (Bjerselius et al. 2000; Li et al. 1995), and model species for toxicology and monitoring (i.e., fathead minnow, *Pimephales promelas*) (Russom et al. 1997; Ankley and Villeneuve 2006). Furthermore, relatively few studies focus on entire fish communities (Poos et al. 2008), scarce species, or those with low perceived importance (Northcote 1998; Lucas and Baras 2001; Knaepkens et al. 2004), despite their potential importance to food webs, biodiversity, and ecosystem services.

Because we know so little about movement of noncommercial and game fishes, we conducted a detailed analysis of movement to (*i*) describe the movement behaviour of a complete community of fishes and (*ii*) test the adequacy of qualitative literature information by testing predictions regarding the habitat use and movement behaviour of stream fishes. Movement data for fishes were collected over 3 years from six adjacent tributaries of Lake Ontario. Over 15 000 individuals from 37 species were captured, including more than 4500 PIT-tagged individuals from 26 species to test four predictions. First, we tested whether arrival times in streams were related to water temperature and then tested whether stream water temperature upon arrival matched estimates of water temperature for arrival in literature accounts. Second, by recording the life stage (juvenile or adult) and sex of individuals, we inferred why individuals were moving into the streams to test the hypothesis that spawning is the primary reason for migration. Third, we compared stream residence times estimated for migrants in our study with general estimates of residence times provided in the literature. Fourth, we compared the proportions of individuals using lake and stream habitats versus those using only stream habitats. While our comparisons do not consider all aspects of movement behaviour, they provide a reasonable test of the utility of some of the literature data on the movement behaviour of fishes that managers might acquire from the literature to lead to better fisheries management and conservation decisions.

Materials and methods

Study sites

Our study was conducted using fishes collected and tracked from late March to late June of 2005–2007 in six adjacent tributaries of Lake Ontario: Cobourg Brook (43°57′40″N, 78°10′39″W), Covert Creek (43°57′35″N, 78°6′25″W), Grafton Creek (43°58′3″N, 78°3′20″W), Shelter Valley Creek (43°57′58″N, 77°59′58″W), Colborne Creek (43°58′49″N, 77°54′1″W), and Salem Creek (43°59′58″N, 77°49′53″W) (Fig. 1). Tributaries were 4.3–8.3 km apart (mean = 5.8 km) when measured from mouth to mouth. All tributaries had in-stream barriers located within 0.4–2.1 km (mean = 0.97 km) of

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the tributary mouth, which is common for Great Lakes tributaries in southern Ontario. Cobourg Brook, Grafton Creek, Shelter Valley Creek, and Colborne Creek have low-head dams (~1.0–1.7 m in height) used to restrict the reproductive migrations of invasive sea lamprey (Porto et al. 1999; Baxter et al. 2003). Covert and Salem creeks have elevated culverts about 1 and 2 m above the stream bed, respectively, with no fishway. Physical and hydraulic characteristics of the five main study streams are summarized in Table A1.

Quantification of timing, size, and sex

Arrival of individuals from various species in each tributary was quantified using nets and monitoring stations for PIT tags. Netting involved daily operation of hoop or trap nets in each tributary except Cobourg Brook. Hoop nets (Murphy and Willis 1996) were placed 80–685 m (mean = 300 m) upstream of the stream mouth and used to sample the entire stream width in Covert (stretched mesh size 2.5 mm), Grafton (stretched mesh size 4 mm), and Salem creeks (stretched mesh size 15 mm) and about 50% and 75% of Colborne (stretched mesh size 4 mm) and Shelter Valley creeks (stretched mesh size 15 mm), respectively. Although the hoop nets in Colborne and Shelter Valley creeks did not cover the entire stream width, they were placed at locations where the noncovered portion of the stream was dominated by intermittently submerged sand bars, where depths were likely too shallow for most fish to pass. Trap nets (Murphy and Willis 1996) (stretched mesh size 12.5 mm) were located 150 m (Shelter Valley Creek) and 170 m (Colborne Creek) upstream of the stream mouth and used in 2005 to supplement the hoop nets. All nets were oriented downstream to capture fish entering the streams from the lake. Differences in mesh sizes are unlikely to have biased our results because even the largest mesh size used was able to catch immature individuals of the smallest species. Nets were placed as close to the mouth of the tributary as possible. However, at Grafton, Shelter Valley, and Colborne creeks, the nets could not be placed right at the mouth because the estuary was too deep for effective netting, whereas at Covert Creek the estuary was difficult to access. Cobourg Brook was not sampled using nets, but was included in the study because a PIT tag detection station from an earlier study (Pratt et al. 2009) detected some of our PIT-tagged fish.

Each day, captured fish were identified to species, examined for state of reproductive maturity and sex, scanned for the presence of a PIT tag using a portable PIT tag reader (Allflex RFID Portable Reader), and measured for fork length to the nearest millimetre. State of maturity and sex of an individual were determined by first squeezing its abdomen for the presence of eggs or milt and then by examining for the presence of conspicuous secondary sexual traits (Table A2), using sexually dimorphic traits described in the literature (Scott and Crossman 1998; Holm et al. 2009). Individuals were classified as unknown gender if they lacked identifiable sexual attributes, but were assessed to be large enough to mature later in the spawning season based on size-at-maturity estimates from the literature (Scott and Crossman 1998). Males, females, and individuals of unknown gender were also referred to as adult individuals in some analyses. Individuals were classified as juveniles if they were smaller than the normal size-at-maturity and displayed no evidence of sexual maturity or secondary sexual traits.

PIT-tagging

Unmarked individuals of all species greater than 100 mm in fork length were PIT-tagged. Individuals were anesthetized in a bath of 0.2 mL·L^{-1} clove oil until loss of equilibrium. A surgical incision was made in the ventral cavity: 4–5 mm off of the midline and just anterior of the pelvic girdle for teleost fishes (Adams et al. 1998); or 1–2 mm off the midline anterior of the gills slits (where the first dorsal fin begins) for sea lamprey. A half-duplex PIT tag (23 mm × 4 mm) (Oregon RFID) was then inserted into the body cavity. The incision was closed using external tissue adhesive (3M Vetbond Tissue Adhesive, 3M, St. Paul, Minnesota). The individual was allowed to recover in a 68 L container filled with fresh stream water and released several metres upstream of the capture point. Loss or shedding of PIT tags was not measured; however, only 15 of 564 (2.7%) individuals recaptured over the course of the 3-year project had an obvious scar at the incision point, but no detectable PIT tag. Tagging mortality was ~1.4%, with 62 dead individuals recovered within 5 days following tagging, comparable to other studies using similar techniques (Sigourney et al. 2005; Bateman and Gresswell 2006). Only one individual died during the ~20–30 min recovery period prior to being released in the streams.

Quantification of movement

Movement of PIT-tagged fishes into, within, and between the study streams was monitored from March to late June of 2005–2007 using two PIT tag detection arrays per stream, which were powered by deep cycle marine batteries exchanged every 7 days, on average. Each array consisted of two antennas placed 2.3–17.4 m apart (mean = 6.7 m), spanning the width of the stream (details of operation in Table A1). We used a paired-antenna design to infer an individual fish's direction of movement from the temporal order of detection by the upstream and downstream antennas at each station. Detections at the downstream stations were also used to infer the timing of immigration and emigration for each species into and from a study stream.

For each stream, the downstream array was positioned 21–240 m from the stream mouth (mean = 110 m). Downstream arrays were positioned as close to the mouths of the streams as possible given the constraints on accessibility and the effects stream width and depth can have on the efficiency of the antennas. The upstream array was positioned just downstream of the first instream barrier to fish movement, 370–2030 m from the stream mouth (mean = 970 m). Arrays recorded the PIT tag number, date, and time a tagged fish was detected passing an antenna. 100% of the stream width was sampled by the antennas. While we may not have tagged all the fish entering the stream, our PIT tag readers could detect movements of all tagged fish within the stream and as they departed the stream.

Stream water temperature

Stream water temperatures were recorded at each of the five main study streams, from late March to late June of 2005–2007, using a HOBO Pendant Temperature/Alarm Data Logger model UA-001-08 (Onset Computer Corporation, Bourne, Massachusetts). Each logger was secured within a 15 cm piece of polyvinyl chloride (PVC) piping, attached to the upstream antennas in flowing water, roughly 5 cm off the bottom, and set to record water temperature at 15 min intervals. Data were downloaded when the antennas were removed at the end of the field season.

Data analysis

Water temperatures at the time of arrival

We tested whether arrival times in streams for the tracked fish were related to spawning migrations as described in the literature. However, spawning times reported in the literature usually covered an extended period (e.g., May–June) or were imprecise (e.g., late spring), so we used water temperatures for spawning reported in the literature as a surrogate for arrival time. This analysis was addressed in two parts. First, we tested whether arrival times were related to water temperature and then related observed water temperatures with those reported in the literature. Arrival date was estimated as the median arrival time of all adults for a species from the telemetry and net-capture data. Next, we determined the water temperature for each yearly sample that corresponded to the median arrival time of the species in a particular stream. Mean water temperature was then calculated as the average of all yearly samples (i.e., n = 18) for a species. For the

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analyses, we included only species with a total of 10 adults in each year of the study to provide a reasonable estimate of water temperature for each species and yet maximize the sample size. In total, 11 415 adults from 18 species were included in the analysis.

Second, we compared the mean water temperature at arrival for the fishes in this study with spawning temperatures from the literature to determine if movements into the streams corresponded to spawning migrations. Water temperature at spawning was obtained from Portt et al. (1988), Scott and Crossman (1998), Holm et al. (2009), Barrett and Munkittrick (2010), and FishBase (http://www.fishbase.org/). Whenever possible, the exact value cited in the literature was used. When a range was given, the midpoint of the range was used. The median value was used when there was no consensus among literature sources.

Reasons for using stream habitat

We used the information collected on reproductive status, sex, and size relative to size at maturity to infer whether individuals were moving into the streams for reproductive or nonreproductive purposes. Nonreproductive purposes, such as foraging or seeking refuge, were suggested by the presence of juvenile fish. Only species with a minimum of 10 adult-sized individuals were included in the analysis to provide a reasonable estimate for each species and yet maximize the sample size. A total of 11 442 individuals from 28 species were used in the final analysis.

Time spent in the streams

The time spent in the streams, quantified only for individuals detected leaving a stream, was the difference between an individual's arrival and departure dates. These values were then aggregated and used to calculate the mean time spent in the streams for each species. Only species with minimum of 10 individuals detected leaving the streams over the 3 years were used to provide a reasonable estimate of time spent in the streams for each species and yet maximize the sample size. A total 1279 adult-sized individuals from 12 species were used in the final analysis. Our estimates of time spent in the streams are likely an underestimate, because it ignored fishes departing after the field season and individuals that remained in the streams. However, our goal was to gain a general sense of the time spent in stream habitats for each species. A two-way ANOVA without replication (Sokal and Rohlf 1969) was used to determine if the time spent in the streams differed among species and among years, with species and year as fixed factors, and mean duration time as the dependent variable.

Literature accounts for time spent in the streams for most species were obtained from Scott and Crossman (1998). Additional information was obtained for lake chub (*Couesius plumbeus*) (Brown et al. 1970), pumpkinseed (*Lepomis gibbosus*) (Danylchuk and Fox 1996), rock bass (*Ambloplites rupestris*) (Gross and Nowell 1980), and smallmouth bass (*Micropterus dolomieu*) (Brown et al. 2009). Whenever possible, the exact value cited in the literature was used. When a range was provided, the midpoint of the range was used.

To determine whether the time spent in streams was related to spawning, we compared our data with those from the literature for each species. Rainbow trout (*Oncorhynchus mykiss*) were excluded from this comparison because many adults had already arrived in the streams prior to net placement, were not tagged, and would have likely biased the results. Also, brook char (*Salvelinus fontinalis*) were excluded from the analysis because this species spawns in the fall, and estimates on time spent in the streams from the literature are typically given for the spawning season. To be included in the analysis, each yearly sample consisted of a minimum of 10 adult-sized individuals to provide a reasonable estimate of time spent in the streams for each species and yet maximize the sample size. Overall, 1121 adult-sized individuals from nine species were analysed.

Relative use of stream and lake habitats

Telemetry data from the antenna arrays were used to determine a species' relative use of stream and lake habitats. For all analyses, the lake habitat starts from the downstream antenna out to the lake. PIT-tagged individuals were assigned to one of four categories: (1) fish with unknown histories — never detected after tagging; (2) residents — detected only in their original stream of capture (by either the upstream or downstream arrays) or recaptured in the nets operated in their original stream of capture and tagging; (3) emigrants — detected by both antennas in the lower array in the correct temporal order or detected on only the downstream antenna of the lower array and never detected or recaptured afterwards; and (4) interstream migrants — detected or recaptured in another stream. Categories were then aggregated for each species to identify individuals that only used the stream habitat (categories 1 and 2) or used both stream and lake habitat (categories 3 and 4). These categories were then used to estimate the proportion of individuals that used the lake. Individuals from category 1 were treated as stream residents that did not move far enough to be detected by the arrays or be recaptured in the nets. A study of stream fish combining similar tagging and tracking techniques with electrofishing surveys and stable isotope analysis determined that individuals with unknown histories were nearly always stream residents (Coppaway 2011). Individuals from categories 3 and 4 were combined because they likely, and definitely, left the streams for Lake Ontario, respectively. Only species with a minimum of 10 individuals (total of adult-sized individuals and juveniles) were used to estimate habitat use in this analysis. Ten individuals were considered adequate to provide a reasonable estimate for each species and maximize the number of species used. Overall, 4864 individuals from 18 species were included in the analysis.

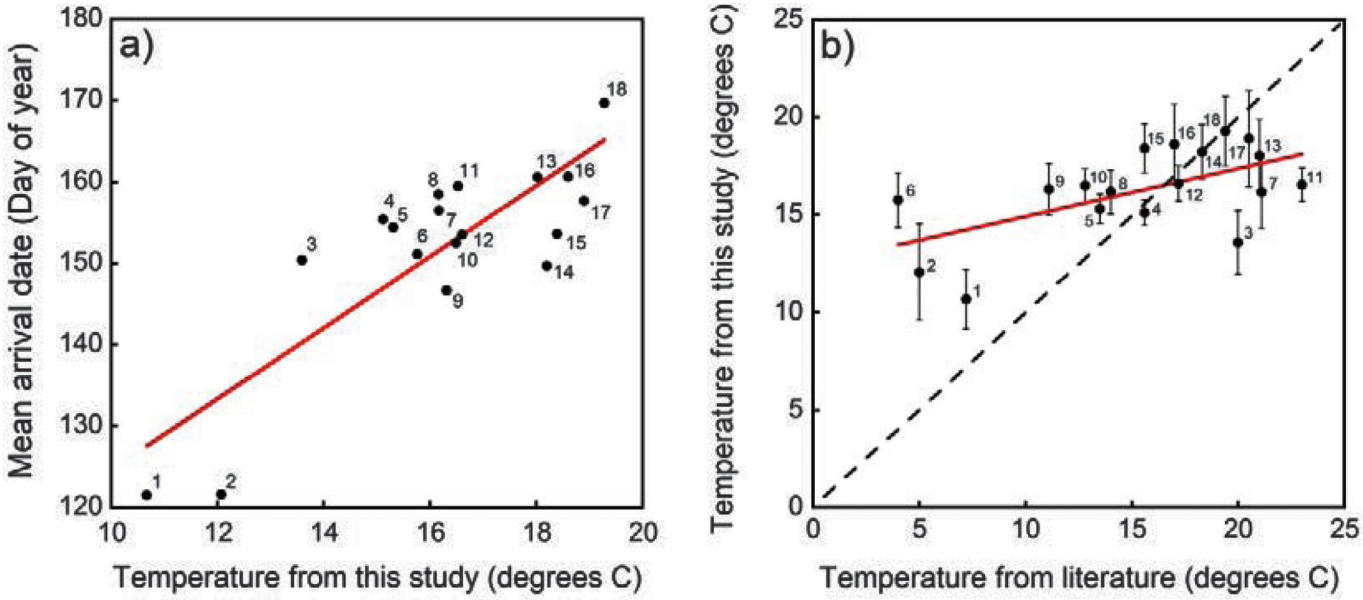
Literature estimates of habitat use were made using information provided from the Fish Migration and Passage Knowledgebase (http://fishmap.uoguelph.ca/). Species were classified as moving between lakes and rivers (i.e., use of lake habitat), moving within rivers (i.e., use of stream habitat), or as having their movement behaviour described as uncertain (Mandrak et al. 2003). A species was listed as uncertain when no information indicated the species moved major distances over its life-span. We treated these species as resident, given the lack of evidence that they move out of streams. These species were then added to the species reported as moving within rivers. Relative use of lake habitat was calculated as the number of studies reporting movements between lakes and rivers divided by the sum of all studies. We recognize that our estimate of use of lake habitat-based individuals moving in and out of tributaries is different than the records of migratory and resident individuals from natural history accounts; however, we considered it valuable from a predictive perspective to test whether the two measures were correlated.

For our comparison, one species (round goby, *Neogobius melanostomus*) was removed because measures of habitat use from the database were lacking and the species is still invading part of the Great Lakes and hence not at equilibrium in terms of habitat occurrence. Sea lamprey was also excluded from the analysis because it is semelparous with adults dying after spawning. All analyses were done using SPSS version 12.0.1, with a critical level of significance set at 0.05.

Results

Overall, 15 375 individuals from 37 species were caught during the study, including 5143 females, 3811 males, 2538 adults of unknown sex, and 3883 juveniles (Table A3). Of these, 4586 individuals from 26 species were PIT-tagged, consisting of 1174 females, 1031 males, 827 adults of unknown sex, and 1554 juveniles.

Fig. 2. Relationship between the mean stream water temperatures at arrival in this study (with 95% CI for panel *b*) and (*a*) median date of arrival of species with a minimum of 10 individuals in each year of the study (n = 18) and (*b*) mean spawning temperatures reported in the literature. Solid line represents least squares line. Dashed line represents a 1:1 line. Species are coded as follows: (1) white sucker; (2) rainbow trout; (3) bluntnose minnow; (4) common shiner; (5) Johnny darter; (6) brook char; (7) northern redbelly dace; (8) lake chub; (9) sea lamprey; (10) creek chub; (11) brown bullhead; (12) longnose dace; (13) blacknose dace; (14) yellow perch; (15) fathead minnow; (16) brook stickleback; (17) rock bass; (18) pumpkinseed.



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Water temperature at the time of arrival

Median arrival times of adults for the 18 species analysed were positively correlated with the mean stream water temperatures (Pearson correlation, r = 0.83, P < 0.001, n = 18), with earlier arrival corresponding with lower water temperatures in the streams (Fig. 2*a*). When only spring-spawning species were considered (excluding rainbow trout and brook char), there was again a significant positive correlation between median arrival date and stream water temperatures (Pearson correlation, r = 0.75, n = 16, P = 0.001).

Stream water temperatures at arrival in this study were also positively correlated with those cited in the literature for spawning. For the 18 species with a minimum of 10 individuals in each year of the study, mean stream water temperature at arrival in this study was significantly correlated with values cited in the literature for water temperatures at spawning (Pearson correlation, r = 0.59, n = 18, P = 0.01; Fig. 2*b*). Similarly, when only spring-spawning species were considered (excluding rainbow trout and brook char), there was again a significant positive correlation between stream water temperature at arrival in this study and stream water temperatures at spawning (Pearson correlation, r = 0.55, n = 16, P = 0.029). We also calculated the slope (0.41) using model II regression (Sokal and Rohlf 1969); the resulting slope still differed from 1 (95% confidence interval (CI) = 0.23–0.59).

Reasons for using stream habitats

Individuals from many species were moving for reasons other than reproduction. A total of 7547 of 15 375 individuals (49.1%) entering the streams were in spawning condition. When only adults were considered (males, females, and individuals of unknown gender), 65.8% of individuals entered the streams in spawning condition, and this number increased to 84.3% when only males and females were considered. Of the 28 species with at least 10 adult-sized individuals captured, 27 spawn in the spring and only one species (brook char) spawns in fall. Only 2 of the 27 spring-spawning species had no individuals in spawning condition, whereas 15 of the 27 had more than 50% in spawning condition (Table 1). All individuals of the fall-spawning species were not in spawning condition (Fig. 3; Table A4).

Individuals in nonspawning condition (unknown sex, adult-sized immature, and juveniles) represented half of all individuals captured (50.1%) for spring-spawning species. Juveniles alone made up ~25% of all individuals captured (n = 3909), and the percentage of individuals in nonspawning condition varied considerably among species. The percentage of individuals in nonspawning condition was high (64.4%–96.8%) in rainbow trout, brown bullhead (*Ameiurus nebulosus*), round goby, yellow perch, emerald shiner (*Notropis atherinoides*), mottled sculpin (*Cottus bairdii*), rock bass, and pumpkinseed, and low (0.82%–37.9%) for sea lamprey, rainbow smelt (*Osmerus mordax mordax*), longnose dace *Rhinichthys cataractae*, northern redbelly dace (*Chrosomus eos*), bluntnose minnow (*Pimephales notatus*), common shiner (*Luxilus cornutus*), golden shiner (*Notemigonus crysoleucas*), blacknose dace (*Rhinichthys atratulus*), and fathead minnow (Table 1).

Time spent in the tributaries

Mean time that individuals spent in the streams differed significantly among the 12 species analysed (two-way ANOVA, species effect: $F_{[11,22]}$ = 2.33, P = 0.044). On average, adults of all species spent 9.3 days in the streams. Durations ranged from 1.3 days for brook char to 23.0 days for smallmouth bass (Table 2). Time spent in the streams did not differ significantly among years (two-way ANOVA, year effect: $F_{[2,22]}$ = 2.77, P = 0.084).

We also tested if the mean time spent in the streams was related to the median arrival date of a species, since the estimated length of time an individual could spend in the stream is likely dependent on the length of time between when an individual was tagged and when the antennas were removed (henceforth referred to as "time at large"). For the 12 species with at least 10 adult-sized individuals, there was no significant correlation between median arrival date and the mean time spent in the streams (Pearson correlation, r = 0.12, n = 12, P = 0.70), suggesting that the time spent in the stream did not differ between individuals tagged shortly before the removal of the antennas compared with individuals tagged early in the season.

Estimated times spent in the streams of adults (rainbow trout and brook char were excluded) corresponded well with estimates

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Table 1. Inferred reasons for movements into the streams based on the proportion of adult-sized individuals in spawning condition captured in streams for each of 30 species (with a minimum of 10 total individuals) and the proportion of juveniles for each species.

Species	Use of streams[a]	Proportion of adult-sized individuals in spawning condition	Proportion juveniles	Total
Banded killifish	Nonreproductive	0.00	0.00	17
Blacknose dace	Reproductive	0.67	0.04	1252
Bluntnose minnow	Reproductive	0.73	0.00	695
Brook char	Nonreproductive	0.00	0.20	116
Brook stickleback	Reproductive	0.55	0.07	354
Brown bullhead	Nonreproductive	0.06	0.34	175
Brown trout	Nonreproductive	0.00	0.94	78
Chinook salmon[b]	Nonreproductive	—	1.00	230
Common shiner	Reproductive	0.72	0.09	314
Creek chub	Reproductive	0.69	0.18	1717
Emerald shiner	Nonreproductive	0.18	0.09	275
Fathead minnow	Reproductive	0.64	0.03	832
Golden shiner	Reproductive	0.68	0.05	119
Johnny darter	Reproductive	0.53	0.11	216
Lake chub	Nonreproductive	0.45	0.00	105
Largemouth bass	Nonreproductive	0.10	0.09	11
Logperch	Reproductive	0.85	0.07	14
Longnose dace	Reproductive	0.79	0.00	2030
Mottled sculpin	Nonreproductive	0.27	0.13	55
Northern redbelly dace	Reproductive	0.77	0.01	1388
Pumpkinseed	Nonreproductive	0.42	0.15	208
Rainbow smelt	Reproductive	0.87	0.00	30
Rainbow trout	Reproductive	0.65	0.95	2141
Rock bass	Nonreproductive	0.29	0.13	389
Round goby	Nonreproductive	0.34	0.87	400
Sea lamprey	Reproductive	0.99	0.00	490
Smallmouth bass	Nonreproductive	0.31	0.00	13
Threespine stickleback	Nonreproductive	0.00	0.00	14
White sucker	Reproductive	0.64	0.35	1497
Yellow perch	Nonreproductive	0.05	0.02	179

[a]Use of streams for reproduction if >50% of adult-sized individuals were sexually mature.
[b]No adults were captured for this species.

provided in the literature (Pearson correlation, r = 0.79, n = 9, P = 0.012; Fig. 4). Time spent in the stream was noticeably underestimated in four species: creek chub, common shiner, brown bullhead, and pumpkinseed; the latter two species were also some of the last to arrive at the streams. We also calculated the slope (1.12) using model II regression (Sokal and Rohlf 1969), which did not differ from 1 (95% CI = 0.50–1.73).

Relative use of lake and stream habitats

For the 18 species with at least 10 individuals of all age classes (n = 4864), there was considerable variation in their relative use of lake habitat, as indicated by the differences in proportion of individuals leaving a stream (categories 3 + 4) versus remaining in the stream (G test: G = 2354.42, df = 17, P < 0.001; Table 3; Fig. 5). Species on the stream-resident end of the spectrum included creek chub (9.1% using the lake), brook char (12.3%), longnose dace (17.0%), brown trout (19.2%), logperch (30.0%), largemouth bass (30.0%), and common shiner (31.9%) (Fig. 5). Species more toward the lake end of the spectrum included brown bullhead (42.4%), pumpkinseed (43.3%), rock bass (49.6%), yellow perch (52.6%), and smallmouth bass (75.0%).

The relative use of stream habitat differed significantly in species known to undertake "anadromous-like" spawning migrations (i.e., moving between lakes and streams) (G test: G = 866.59, df = 5, P < 0.001). For example, rainbow smelt (10.5% use of lake habitat) and rainbow trout (35.5%) were on the stream-resident end of the continuum, whereas lake chub (73.7%) were detected more often leaving for the lake. White sucker were observed to use both stream and lake habitat almost equally (47.6%). Differences between the relative habitat use estimated in this study compared with the expected based on life history could possibly indicate that these populations exhibit partial migration.

We compared the relative use of lake habitat in the literature (Table 4) with those reported in this study (i.e., categories 3 + 4). Percent use of lake habitat in this study did not correlate with that reported in the literature (Pearson correlation, r = 0.04, n = 16, P = 0.89), nor when known "anadromous" species were removed from the analysis (Pearson correlation, r = 0.05, n = 12, P = 0.88; Table 4; Fig. 6).

Discussion

Our findings indicate that fishery managers can use natural history information to gain general insights into the movement ecology of freshwater fishes, but should also recognize that this information remains incomplete in important ways. Our conclusion regarding general insights is supported by the analyses demonstrating that literature data on the timing of arrival to the streams, water temperature at arrival, and the time spent in the streams were reasonably good predictors of the patterns in our data. Further, fish may move into the streams to reproduce, forage, or seek refuge from predators or less hospitable environmental conditions (Northcote 1997; Lucas and Baras 2001). Our observations indicated that the majority of the fish moving into the streams were in spawning condition and therefore arriving for reproductive purposes, although exceptions clearly existed (see below). Further, although not presented in this paper, this timing was also consistent across years, illustrating the seasonal and temporal predictability of these movements into the streams for these species (Hendry et al. 2004).

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Fig. 3. Proportion of immature adult-sized individuals, summarized by species with a minimum of 10 adult-sized individuals. The dashed line represents an equal proportion (0.50) of mature and immature individuals. Species are coded as follows: brook char (BT), banded killifish (BKF), threespine stickleback (3SSB), yellow perch (YP), brown bullhead (BBH), largemouth bass (LMB), emerald shiner (ES), mottled sculpin (MS), rock bass (RB), smallmouth bass (SMB), round goby (GOBY), pumpkinseed (PMKS), lake chub (LC), Johnny darter (JD), brook stickleback (BSTB), white sucker (WS), fathead minnow (FHM), rainbow trout (RBT), blacknose dace (BND), golden shiner (GS), creek chub (CC), common shiner (CS), bluntnose minnow (BLNM), northern redbelly dace (NRBD), longnose dace (LND), logperch (LGP), rainbow smelt (RBS), and sea lamprey (SL).

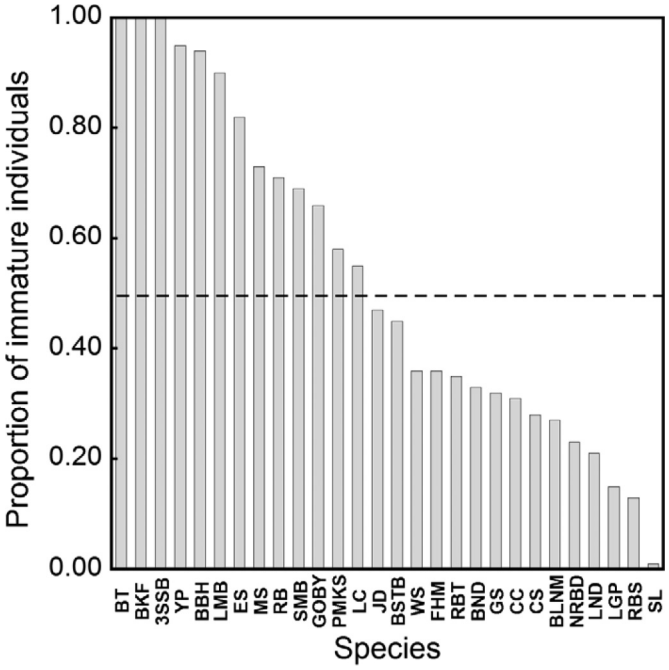
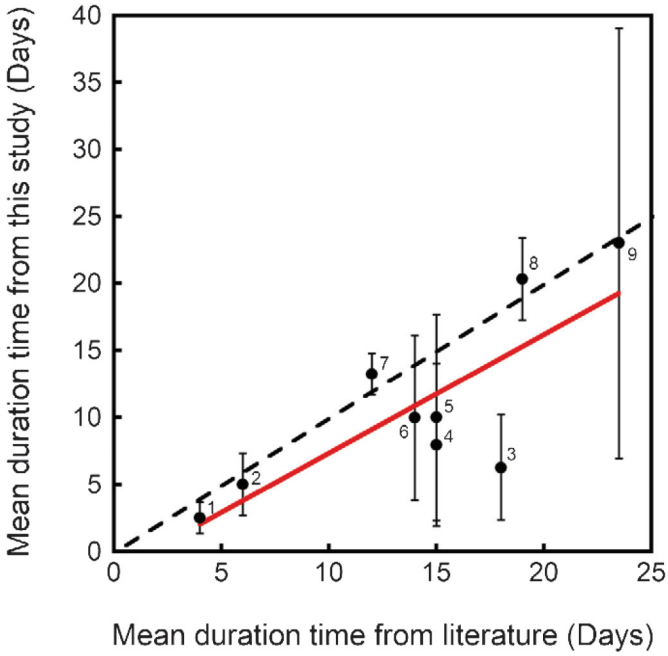


Fig. 4. Relationship between the time spent in the streams reported in the literature against the time spent in the streams estimated in this study for species (n = 9) with a minimum of 10 individuals detected leaving the streams (with 95% CI). Solid line represents least squares line. Dashed line represents a 1:1 line. Species are coded as follows: (1) sea lamprey; (2) lake chub; (3) brown bullhead; (4) common shiner; (5) pumpkinseed; (6) creek chub; (7) white sucker; (8) rock bass; (9) smallmouth bass.



Our conclusion regarding potential limitations is partly supported by two novel findings. First, in many species juveniles were also moving into streams, even in those species where adults were clearly spawning in the streams. This finding is noteworthy because most natural history sources make little reference to the movements of juvenile fish. It is also noteworthy because it indicates that fish were moving into the streams for reasons other than reproduction. Streams can be important habitats for the juvenile age classes. They provide productive habitats where individuals can grow fast, owing to warmer temperatures in streams relative to lakes (Kishi et al. 2005; Bal et al. 2011), as well as seek refuge from larger predators found in lakes (Northcote 1997; Lucas and Baras 2001; Salas and Snyder 2010). In streams, foraging and refuge habitats can also be similar in habitat structure and location (Lucas and Baras 2001). Juveniles might also move into streams to acquire information about spawning locations and behaviour that they will use later in life (Dodson 1988; Lucas and Baras 2001). Whatever the reason for juveniles entering the streams, we believe that the evidence regarding the movement of juvenile fish in this study is important and is rarely studied for freshwater fish. The magnitude of juvenile movements, in terms of number of species, individuals within species, and across streams and years, suggest that these movements are important ecologically and worthy of further research.

Second, adult-sized individuals were moving into streams for nonreproductive purposes. This observation was clearest for the fall-spawning brook char and brown trout, but also for brown bullhead, yellow perch, pumpkinseed, lake chub, rock bass, mottled sculpin, emerald shiner, and banded killifish. The latter five species are of particular note, because small-sized fish are rarely observed in studies of migratory behaviour.

There were also features in the patterns we detected that suggested data limitations arising from differences in ecology and life history among species and to the timing of our study. For example, the slope and intercept of our comparison of water temperature at arrival differed significantly from a 1:1 relationship. The deviations are likely due in part to unique species differences. Brook char were included in the analysis of water temperatures upon arrival; however, they spawn in the fall, so spring water temperatures measured for when they arrived in this study are likely not representative of water temperatures during fall spawning (Fig. 2*b*). Similarly, our data suggested that individual brown bullhead were leaving the streams significantly earlier than cited in the literature (Fig. 4). Interpretation of these data are complicated because males provide parental care and would be expected to remain longer in the streams (Scott and Crossman 1998). Also, our estimated water temperatures at arrival for rainbow trout were warmer than those cited in the literature (Fig. 2*b*). In our study, adult rainbow trout had likely arrived in the streams prior to when we could set up our field operations (ice-out) each season. These examples emphasize how differences between among, and possibly populations within species, as well as differences in study design and effort (ours and earlier studies) can create variability in data quality and complicate the extrapolation of literature data. One final possibility for the apparent mismatch observed in our comparison of water temperature at arrival may be due to climate change and its effects on stream temperatures (Ficke et al. 2007). It is possible that complex temperature–time of day effects are being affected by climate change and could lead to apparent data

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Table 2. Median date of arrival at, departure from, and duration (time spent) in the study streams by adult-sized (males, females, and individuals of unknown gender) individuals summarized by species and year, in order of median arrival date, as well as the overall median arrival date, departure date, and duration of all adult-sized individuals and juveniles (Juv.) for each species.

	Arrival date					Departure date					Duration (days)	
	Year			Overall median		Year			Overall median		Overall mean	
Species	2005	2006	2007	Adults	Juv.	2005	2006	2007	Adults	Juv.	Adults	Juv.
White sucker	138.4	108.5	116.9	121.5	143.7	171.9	128.1	120.0	143.3	170.2	13.3	10.6
Rainbow trout	136.4	110.7	123.5	121.6	139.5	148.9	120.1	136.0	130.9	150.9	3.4	4.1
Emerald shiner	157.0	147.4	140.6	141.8	156.7							
Sea lamprey	160.2	137.5	154.0	146.7		167.8	141.7	158.0	147.7		2.5	
Yellow perch	161.5	145.5	161.6	149.7	172.4	178.9	135.9	145.7	165.2		8.5	
Bluntnose minnow	156.7	152.4	149.4	150.4	142.6							
Brook char	153.8	135.4	151.6	151.1	153.5	163.2	134.4	157.0	156.1	154.2	1.3	0.3
Creek chub	149.5	153.6	153.5	152.5	157.6	161.6	158.9	153.8	156.6		10.0	
Longnose dace	160.6	151.4	142.6	153.5	150.1							
Fathead minnow	139.0	146.4	155.5	153.6	170.5							
Johnny darter	153.9	150.4	161.0	154.4	149.6							
Common shiner	155.5	155.4	154.7	155.4	157.4	165.0	168.9	164.9	166.0		7.9	
Lake chub	159.7	152.3	151.0	156.5		160.9	157.1	158.4	160.4		5.0	
Rock bass	160.5	151.4	153.4	157.7	131.6	177.1	186.2	173.9	177.9		20.1	
Northern redbelly dace	157.6	167.4	154.6	158.5	157.6							
Brown bullhead	160.7	150.0	141.5	159.5		164.9	173.7	160.9	164.9		6.2	27.0
Blacknose dace	155.6	165.4	161.5	160.6	132.5							
Brook stickleback	154.4	154.1	172.4	160.6	156.5							
Pumpkinseed	171.5	167.7	170.6	169.7	170.6	170.7	180.4	189.3	177.6		10.0	
Smallmouth bass[a]						159.1	157.1	181.3	168.6		23.0	

[a]Smallmouth bass were not included in the analysis of arrival times because fewer than 10 individuals were captured in each year of the study, but were included in the analyses of time spent in the streams (duration) because more than 10 individuals were detected leaving the streams.

Table 3. The movement behaviour of all PIT-tagged individuals (n = 4888), including individuals detected on the antennas in subsequent years, summarized by species, gender, where they were detected, and the percentage of all individuals detected leaving the streams.

	No. of individuals detected per category (1–4)																
	Adult-sized																
	Female				Male				Unknown				Juveniles				
Species	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	Left stream
Blacknose dace									1		1						—
Brook char					1	3			11	66	7	5	4	8		1	12.3%
Brown bullhead		1	5	1					47	19	44	4	13	19	18	1	42.4%
Brown trout										4	3		17	21	7		19.2%
Central stoneroller	1																—
Chinook salmon													5	2	1		—
Common shiner	1	3	2	2	17	28	15	4									31.9%
Creek chub	233	202	31	1	70	120	31			1							9.1%
Emerald shiner							1				2						—
Golden shiner			2							1	2						—
Hornyhead chub					1												—
Lake chub	9	5	30	2	2	9	38										73.7%
Largemouth bass		1								6	3						30.0%
Logperch	2	1	2		2	1				1	1						30.0%
Longnose dace	14	17	6	1	2	3	1		2	1							17.0%
Mottled sculpin										1							—
Northern pike										1	1		1				—
Pumpkinseed	4	3	3			3	8	4	22	6	9	5					43.3%
Rainbow smelt	7				6		2		4								10.5%
Rainbow trout	5	3	9	1	9	13	25	3	8	4	8	2	429	359	381	28	35.5%
Rock bass	5	28	41	4	4	12	22	5	65	97	128	8					49.6%
Round goby	1	4	2			1				2							20.0%
Sea lamprey	55	76	61	3	71	123	83	11	3		1						32.6%
Smallmouth bass		1	1			1	1		1	2	3	10					75.0%
White sucker	108	80	156	35	79	84	139	25	50	66	160	3	135	60	82	2	47.6%
Yellow perch							4	1	10	8	15						52.6%
Total	445	425	351	50	264	401	370	53	224	286	388	37	604	469	489	32	36.2%

Note: Percentage of fish leaving a stream was calculated only for species with at least 10 individuals. Detection categories were as follows: 1, never detected after tagging; 2, detected only in stream of capture; 3, detected leaving a stream; and 4, moved to another stream.

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Table 4. The number of studies reporting on the movement behaviour of fishes between lakes and rivers (use of lake habitat), within rivers only (use of stream habitat), and where movement behaviour was uncertain, the percentage of studies categorizing movement behaviour as uncertain (% uncertain) (Migration and Passage Knowledge Database; Mandrak et al. 2003; http://fishmap.uoguelph.ca/), for species with at least 10 individuals in this study, the proportion of literature studies citing the use of lake habitat and the proportion of individuals detected leaving the streams from this study (categories 3 and 4).

	Movement behaviour					
Common Name	Between lakes and rivers	Within rivers	Uncertain[a]	% Uncertain	Lake use (literature)	Lake use (this study)
Alewife	15	0	0	0	—	—
American brook lamprey[b]	3	2	2	29	—	—
Banded killifish[b]	0	0	6	100	—	—
Blacknose dace[b]	0	4	3	43	—	—
Bluntnose minnow[b]	2	1	4	57	—	—
Brook stickleback[b]	1	4	3	38	—	—
Brook char	4	5	1	10	0.40	0.13
Brown bullhead	3	0	5	63	0.38	0.43
Brown trout	3	1	2	33	0.50	0.19
Central stoneroller	0	3	3	50	—	—
Chinook salmon	12	1	1	7	—	—
Coho salmon	9	0	0	0	—	—
Common shiner	0	5	3	38	0.00	0.32
Creek chub	1	5	3	33	0.11	0.09
Emerald shiner[b]	1	2	3	50	—	—
Fantail darter[b]	0	4	3	43	—	—
Fathead minnow[b]	1	3	5	56	—	—
Golden shiner[b]	3	0	6	67	—	—
Hornyhead chub	0	1	4	80	—	—
Johnny darter[b]	0	2	5	71	—	—
Lake chub	3	0	2	40	0.60	0.74
Largemouth bass	2	0	4	67	0.33	0.30
Logperch	0	1	4	80	0.00	0.30
Longnose dace[b]	2	1	3	50	0.33	0.19
Mottled sculpin[b]	0	2	3	60	—	—
Northern pike	5	0	2	29	—	—
Northern redbelly dace[b]	0	0	3	100	—	—
Pumpkinseed	0	2	6	75	0.00	0.43
Rainbow smelt	7	0	0	0	1.00	0.11
Rainbow trout	10	1	1	8	0.83	0.36
Rock bass	1	3	4	50	0.13	0.50
Round goby[c]	—	—	—	—	NA	0.20
Sea lamprey	9	1	0	0	0.90	0.33
Smallmouth bass	3	5	3	27	0.27	0.75
Threespine stickleback[b]	3	0	2	40	—	—
White sucker	11	5	1	6	0.65	0.48
Yellow perch	4	1	4	44	0.44	0.53

[a]Studies were categorized as uncertain if there was ambiguity regarding the movement behaviour of a species (Mandrak et al. 2003).
[b]Species categorized as small-bodied, with mean fork length of all individuals <75 mm.
[c]No studies were available for the round goby in the database.

mismatches between observations in the field and life history literature.

We did not find a significant relationship in our analysis of the relative use stream habitat. This outcome was most likely due to differences in sampling design and method of measurement between our study and the literature information available. Natural history literature are often compared with current surveys to assess, for example, changes in the biological community, as well as to identify areas of high endemism and biodiversity, which can introduce biases and lead to inappropriate management or conservation actions, particularly when different sampling protocols are used (Smith and Jones 2007). For example, information on habitat use from the literature is often based on studies reporting the occurrence of a species (Smith and Jones 2007), whereas data from this study on the relative use of lake habitat were based on the proportion of individuals leaving the streams for the lake. The relative use of lake habitat for the sea lamprey provide a perfect example of this bias, with proportionately more studies from the literature citing the use of lake habitat (0.90) than from this study (0.33). This is not surprising given that juveniles of this species are known to leave streams for the lake to grow before returning to the streams to spawn (literature accounts of occurrence), while adults die shortly after spawning; hence, few adult individuals would be expected to leave the streams (this study; Beamish 1980). Similarly, of all species in this study, smallmouth bass spent the most time in the streams, yet were classified as using lake habitat. Although these results seem contradictory, they do highlight the biology of the species reasonably well; this species is known to provide parental care (increased time spent in streams) before leaving for lake habitat, highlighted by most individuals detected leaving the streams (Scott and Crossman 1998). Also, since the habitat occurrence of many invasive species has not yet reached equilibrium (e.g., round goby in the Great Lakes), the use of expert opinion rather than literature information may be more appropriate in deducing habitat use of invading species in those regions. Thus, fisheries managers should not dismiss biological

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Fig. 5. Proportion of PIT-tagged individuals never detected (light grey), detected only in their original stream of capture (open bar), detected leaving a stream (dark grey), and detected moving to another stream (black bar), summarized by species with at least 10 individuals. The dashed line represents an equal proportion (0.50) of individuals detected in the stream only and detected entering the lake. Species are coded as follows: smallmouth bass (SMB), lake chub (LC), yellow perch (YP), rock bass (RB), white sucker (WS), pumpkinseed (PMKS), brown bullhead (BBH), rainbow trout (RBT), sea lamprey (SL), common shiner (CS), largemouth bass (LMB), logperch (LGP), round goby (GOBY), brown trout (BNT), longnose dace (LND), brook char (BT), rainbow smelt (RBS), and creek chub (CC).

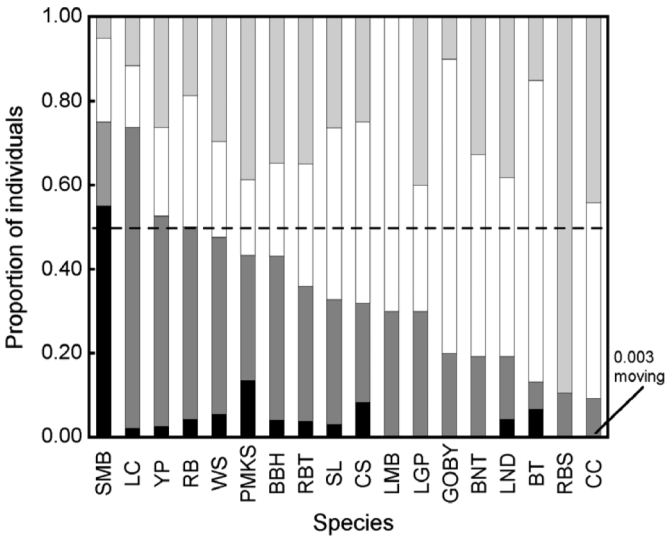
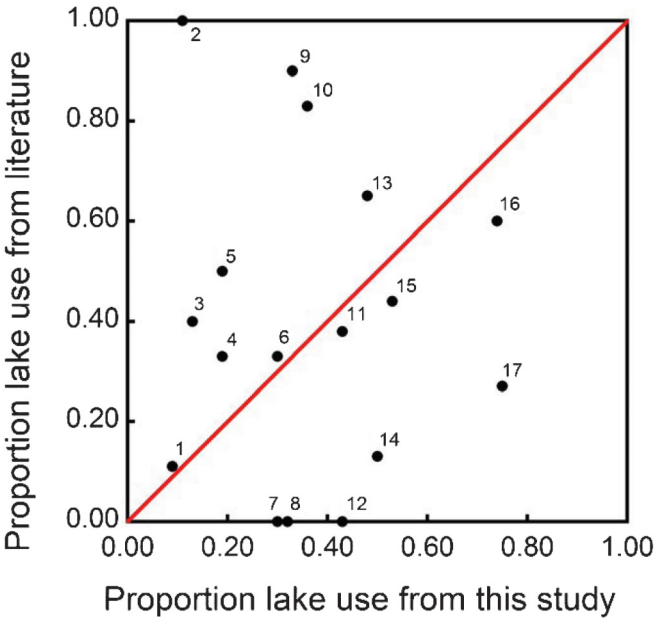


Fig. 6. Relationship between the proportion of individuals using lake habitat reported in the literature versus the actual values of habitat use reported in this study (n = 17). Solid line represents a 1:1 line. Species are coded as follows: (1) creek chub; (2) rainbow smelt; (3) brook char; (4) longnose dace; (5) brown trout; (6) largemouth bass; (7) logperch; (8) common shiner; (9) sea lamprey; (10) rainbow trout; (11) brown bullhead; (12) pumpkinseed; (13) white sucker; (14) rock bass; (15) yellow perch; (16) lake chub; and (17) smallmouth bass.



information simply because of possible uncertainties (Peterman 2004), but the limitations and biases of comparing natural history data with field studies should be made clear, illustrating the need to interpret such comparisons with caution (Smith and Jones 2007).

Our findings are potentially valuable to fishery managers in at least two ways. First, analyses of the timing of movements can be used to help schedule construction activities in and near streams in ways that minimize their effects on fish. These activities could include construction and maintenance of road crossings, stabilization of banks, and the construction and removal of dams. Additionally, extending the observations to other times of year would provide valuable information regarding fish movements other than during the spawning season. Our findings regarding the movements of juveniles could also be important for management plans involving fish passage and the impacts of obstructions to fish movement. Typically these decision consider the upstream movement of adults and downstream movement of juveniles past dams. Our observations of juveniles moving into streams suggest that upstream movement of juveniles may be important in terms of understanding the effects of stream fragmentation, yet are rarely considered in fish passage decisions. Juveniles may be at greater risk relative to adults, since their jumping and swimming abilities could be more limited because of their smaller body size; hence, they could also be more vulnerable to predators near dams or obstructions.

A strength of our study is that we have tracked small fishes that are typically ignored in fish movement studies. Although the natural history data on timing and nature of movements are representative for larger species, they remain to be assessed, for the most part, for small-bodied fishes. Few field studies have examined the movement behaviour and use of habitat in small-bodied freshwater fishes (Bruyndoncx et al. 2002; Cookingham and Ruetz 2008; Breen et al. 2009), with most studies focusing on species with commercial or recreational importance (i.e., salmonids; Ombredane et al. 1998; Sigourney et al. 2005; Bateman and Gresswell 2006). Studies on smaller-bodied fishes often preclude the use of electronic tags (i.e. PIT tags), which can adversely affect tagging mortality in smaller individuals (Roussel et al. 2000), limiting both the types and numbers of fish that could be followed. Indeed, of all fishes categorized as small-bodied, we were able to gather information on the habitat use and movement behaviour of only one species, the longnose dace (Table 4). Improvements in tagging technologies, as well as the recognition of the importance of smaller-bodied fishes to biodiversity and ecosystem services, could potentially increase the number and type of fishes that could be tracked and provide novel insights regarding the movement behaviour and use of habitat in these smaller-bodied freshwater fishes (Roussel et al. 2000; Fischer et al. 2001).

Our analyses are important for the conservation and management planning of stream fishes. Scientists believe that the quality of conservation or recovery plans can be improved by the addition of more biological information (Abell 2002; Poos et al. 2008). However, fishery managers routinely have to make decision with imperfect information and limiting funding and time for new research (Smith and Jones 2007). In such cases, the use of literature data can be an important source of planning information. The ability to implement conservation plans, and their success, can also be limited if information regarding life history and movement behaviour is anecdotal, qualitative, or simply unavailable (Mandrak et al. 2003; Smith and Jones 2007). Assessments of historical data can help validate their utility, but it is also important to assess the quality limits of existing data to ensure that data are used most effectively. This is especially true for movements of stream fishes where sampling biases have been revealed (Gowan and Fausch 1996) and tracking technologies have improved dramatically. Results from this study suggest that natural history

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information is a useful tool for key features of the movement behaviour of freshwater fishes when new studies are not feasible owing to funding or time demands. However, it also reveals limitations and biases arising from differences between historical and current studies in sampling protocols and effort and the ability to track fish (Smith and Jones 2007).

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Appendix

Table A1. Summary of physical features and sampling and resampling effort for each stream and year of the project.

	Stream														
	Covert			Grafton			Shelter Valley			Colborne			Salem		
Variables	2005	2006	2007	2005	2006	2007	2005	2006	2007	2005	2006	2007	2005	2006	2007
Stream feature															
Discharge ($m^3 \cdot s^{-1}$)	0.09	0.29	0.17	0.16	0.68	0.39	0.62	1.52	1.14	0.31	1.05	0.73	0.16	0.31	0.22
Width (m)	3.71	3.70	3.40	4.08	4.36	4.15	9.33	9.49	9.05	5.87	6.32	6.14	6.09	6.10	6.19
Depth (m)	0.17	0.27	0.22	0.20	0.30	0.25	0.26	0.38	0.40	0.27	0.39	0.37	0.54[a]	0.32	0.28
Velocity ($m \cdot s^{-1}$)	0.14	0.29	0.23	0.20	0.52	0.38	0.26	0.42	0.31	0.20	0.43	0.32	0.05	0.16	0.13
Water temp. (°C)	15.5	14.1	14.1	15.1	14.2	14.1	16.4	15.0		17.8	16.5	16.5	14.9	15.4	15.3
Distance to barrier (km)	1.1			0.4			0.6			0.9			2.1		
Sediment type															
Lower reach	Bedrock			Silt–sand			Silt			Silt			Silt		
Upper reach	Gravel			Gravel–cobble			Sand–gravel			Bedrock–gravel			Sand–gravel		
Tagging effort															
Start[b]	104	102	121	97	102	121	102	93	121	98	102	121	97	102	121
Netting (days)	53	67	30	59	57	29	84[c]	76	32	81[c]	70	30	61	66	20
Fish caught	1256	1071	768	421	927	1180	631	543	627	432	1262	933	260	647	132
Fish tagged	438	149	128	168	103	105	458	407	166	197	262	159	243	340	49
Detection effort															
Upper (start date)[b]	106	85	115	123	83	110	123	86	107	126	83	111	117	86	111
Effort (days)	72	95	63	57	97	69	56	163[e]	100[f]	53	163[e]	65	61	94	69
Lower (start date)[b]	129	83	114	124	83	107	127	85	104	106	84	104	124	86	104
Effort (days)	49	97	64	56	97	72	62	161[e]	107[f]	139[d]	170[e]	121[f]	121[d]	169[e]	103[f]

[a]The presence of several beaver dams in the study reach likely affected the flow regime for Salem Creek in 2005.
[b]Day of year.
[c]Two nets were used.
[d]Antennas in place until September 2005.
[e]Antennas in place until September 2006.
[f]Antennas in place until August 2007.

Table A2. Secondary sexual traits used to identify males (and females when noted) of the 37 species captured in this study.

Species	Scientific name	Secondary sexual trait
Alewife	*Alosa pseudoharengus*	—
Banded killifish	*Fundulus diaphanus*	Intensive bluish-green colouration (males), females are pale
Blacknose dace	*Rhinichthys atratulus*	Rust-red colouration on sides of body
Bluntnose minnow	*Pimephales notatus*	Nuptial tubercles on snout
Brook char	*Salvelinus fontinalis*	Pronounced kype (hook) on lower jaw
Brook lamprey	*Lethenteron appendix*	—
Brook stickleback	*Culaea inconstans*	Jet black body and fins (sometimes tinged with copper), faint reddish colour on pelvic fins
Brown bullhead	*Ameiurus nebulosus*	—
Brown trout	*Salmo trutta*	Pronounced kype (hook) on lower jaw
Central stoneroller	*Campostoma anomalum*	Nuptial tubercles over most of the body, orange fins
Chinook salmon	*Oncorhynchus tshawytscha*	Pronounced kype (hook) on lower jaw
Coho salmon	*Oncorhynchus kisutch*	Pronounced kype (hook) on lower jaw, brilliant red sides
Common shiner	*Luxilus cornutus*	Nuptial tubercles on snout
Creek chub	*Semotilus atromaculatus*	Nuptial tubercles on snout
Emerald shiner	*Notropis atherinoides*	Nuptial tubercles on pectoral fins
Fantail darter	*Etheostoma flabellare*	Dorsal spines tipped with yellow or orange fleshy knobs
Fathead minnow	*Pimephales promelas*	Nuptial tubercles on snout
Golden shiner	*Notemigonus crysoleucas*	Nuptial tubercles on head, body, and all fins
Hornyhead chub	*Nocomis biguttatus*	Nuptial tubercles covering the entire head, red spot behind the eye
Johnny darter	*Etheostoma nigrum*	Black anterior half of the body
Lake chub	*Couesius plumbeus*	Red mark at the base of the pectoral fin
Largemouth bass	*Micropterus salmoides*	—
Logperch	*Percina caprodes*	—
Longnose dace	*Rhinichthys cataractae*	Orange–red on the corners of the mouth, tips of the pectoral, pelvic, or anal fins
Mottled sculpin	*Cottus bairdii*	Dark band with broad orange distal edge on first dorsal fin
Northern pike	*Esox lucius*	—
Northern redbelly dace	*Chrosomuseos*	Flanks brilliant red below the midlateral band
Pumpkinseed	*Lepomis gibbosus*	Blue–orange color on operculum
Rainbow smelt	*Osmerus mordax mordax*	Small nuptial tubercles on head, body, and fins
Rainbow trout	*Oncorhynchus mykiss*	Pronounced kype (hook) on lower jaw
Rock bass	*Ambloplites rupestris*	Margins of the pelvic and anal fins are black (males) or yellowish-white (females)
Round goby	*Neogobius melanostomus*	Charcoal black, white or yellow edge on dorsal or tail fin
Sea lamprey	*Petromyzon marinus*	Prominent ridge on dorsal sides of body
Smallmouth bass	*Micropterus dolomieu*	—
Threespine stickleback	*Gasterosteus aculeatus*	Blue eyes, red sides and belly
White sucker	*Catostomus commersonii*	Nuptial tubercles on anal and caudal fins
Yellow perch	*Perca flavescens*	Lower fins suffused with orange to bright red

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Table A3. The number of individuals captured in the streams during the study summarized by species, gender, and total number per species, with the number of individuals PIT-tagged in parentheses.

Species	Female	Male	Unknown	Juvenile	Total
Alewife			5		5
Banded killifish[a]			17		17
Blacknose dace[a]	573	354	271 (2)	54	1252 (2)
Bluntnose minnow[a]	400	265	28	2	695
Brook lamprey[a]	2		6		8
Brook stickleback[a]	106	80	144	24	354
Brook trout		4 (4)	89 (86)	23 (13)	116 (103)
Brown bullhead	7 (7)		109 (109)	59 (51)	175 (167)
Brown trout			5 (4)	73 (45)	78 (49)
Central stoneroller	1 (1)				1 (1)
Chinook salmon				230 (8)	230 (8)
Coho salmon				1	1
Common shiner	196 (7)	79 (64)	11	28	314 (71)
Creek chub	1121 (459)	226 (220)	63	307	1717 (679)
Emerald shiner[a]	28	24 (1)	198 (2)	25	275 (3)
Fantail darter[a]		1	1		2
Fathead minnow[a]	446	245	115	26	832
Golden shiner[a]	35 (2)	48	30 (3)	6	119 (5)
Hornyhead chub		1 (1)			1 (1)
Johnny darter[a]	67	44	82	23	216
Lake chub	49 (42)	52 (39)	4		105 (81)
Largemouth bass	1 (1)		9 (9)	1	11 (10)
Logperch	8 (5)	3 (3)	2 (2)	1	14 (10)
Longnose dace[a]	686 (34)	1117 (5)	217 (3)	10	2030 (42)
Mottled sculpin[a]	5	8	35 (1)	7	55 (1)
Northern pike			2 (2)	1 (1)	3 (3)
Northern redbelly dace[a]	700	448	225	15	1388
Pumpkinseed	47 (11)	46 (17)	84 (38)	31	208 (66)
Rainbow smelt	10 (7)	16 (8)	4 (4)		30 (19)
Rainbow trout	24 (17)	58 (44)	24 (22)	2035 (1156)	2141 (1239)
Rock bass	66 (64)	36 (34)	238 (224)	49	389 (322)
Round goby	16 (6)	12 (1)	25 (2)	347	400 (9)
Sea lamprey	197 (195)	289 (288)	4 (4)		490 (487)
Smallmouth bass	2 (2)	2 (2)	9 (9)		13 (13)
Threespine stickleback[a]			14		14
White sucker	349 (313)	345 (294)	275 (265)	528 (280)	1497 (1152)
Yellow perch	1	8 (5)	167 (36)	3	179 (41)
Total	5143 (1173)	3811 (1030)	2512 (827)	3909 (1554)	15 375 (4584)

[a]Species categorized as small-bodied, with mean fork length of all individuals <75 mm. Brook lamprey were included as small-bodied fishes because their narrow body cavity prevented the use of PIT tags.

Table A4. The number of individuals captured during the study summarized by species and gender, with the number of mature individuals in parentheses, as well as the overall proportion of immature individuals (Immature) for species with at least 20 individuals.

Species	Female	Male	Unknown	Juvenile	Immature
Alewife			5		—
Banded killifish			17		—
Blacknose dace	573 (476)	354 (332)	271	54	0.35
Bluntnose minnow	400 (258)	265 (245)	28	2	0.28
Brook lamprey	2 (2)		6		—
Brook stickleback	106 (106)	80 (77)	144	24	0.48
Brook char		4	89	23	1.00
Brown bullhead	7 (7)		109	59	0.96
Brown trout			5	73	1.00
Central stoneroller	1 (1)				—
Chinook salmon				230	1.00
Coho salmon				1	—
Common shiner	196 (139)	79 (67)	11	28	0.34
Creek chub	1121 (750)	226 (216)	63	307	0.44
Emerald shiner	28 (21)	24 (24)	198	25	0.84
Fantail darter		1 (1)	1		—
Fathead minnow	446 (337)	245 (180)	115	26	0.38
Golden shiner	35 (35)	48 (42)	30	6	0.35
Hornyhead chub		1 (1)			—
Johnny darter	67 (58)	44 (44)	82	23	0.53
Lake chub	49 (24)	52 (23)	4		0.55
Largemouth bass	1 (1)		9	1	—
Logperch	8 (8)	3 (3)	2	1	—
Longnose dace	686 (641)	1117 (956)	217	10	0.21
Mottled sculpin	5 (5)	8 (8)	35	7	0.76
Northern pike			2	1	—
Northern redbelly dace	700 (647)	448 (409)	225	15	0.24
Pumpkinseed	47 (40)	46 (34)	84	31	0.64
Rainbow smelt	10 (10)	16 (16)	4		0.13
Rainbow trout	24 (20)	58 (49)	24	2035	0.97
Rock bass	66 (65)	36 (34)	238	49	0.75
Round goby	16 (11)	12 (7)	25	347	0.96
Sea lamprey	197 (197)	289 (289)	4		0.01
Smallmouth bass	2 (2)	2 (2)	9		—
Threespine stickleback			14		—
White sucker	349 (320)	345 (298)	275	528	0.59
Yellow perch	1	8	167	3	0.95
Total	5143 (4182)	3811 (3365)	2512	3909	0.51

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